COURSE HANDOUT
B.Tech- 2nd Semester

Course Title : **Electrical Circuit Analysis-II** Dated: **05-08-2024**
Course Code : **23EE303** Academic Year : **2024-25**
Course coordinator : **Dr.P.Ramana** Course Structure: **3-0-0-3**
Instructor(s) : **Dr.P.Ramana, Dr.G.Chandra Sekhar**

Course Description:
This course deals with 3-phase electric circuits, Transient analysis, Laplace transforms and their application in electrical circuits, different two port network parameters and their interrelationships, network topology and network functions.

Course Objectives:
This course enables the students to:
- Understand the 3-phase balanced and unbalanced circuits
- Understand Laplace transforms and their applications in electrical circuits
- Understand transients and their analysis in electrical circuits
- Understand different two-port network parameters and their inter relationships.
- Understand the concept of network topology
- Outline various network functions.

Textbook (s):
1. W.H.Hayt, J.E.Kimmerly, and S.M.Durbin, "Engineering Circuit Analysis", Tata McGraw Hill, 8th Edition, 2012.
2. Charles K Alexander and Mathew N.O Sadiku, "Fundamentals of Electric Circuits", Tata McGraw Hill, 5th Edition, 2013.
3. Ramana Pilla, "Network Analysis and Synthesis", Universities Press India Pvt. Ltd, 1st Edition, 2018.

Reference (s):
1. M.E Van Valkenburg, "Network Analysis", Prentice Hall of India, 3rd Edition, 2020.
2. Abhijit Chakrabarti, "Circuit Theory Analysis and Synthesis", Dhanpat Rai & Co., 6th Edition, 2014.
3. M Nahvi, Joseph Edminister, K Uma Rao, "Electric Circuits, (Schaum's Outline Series)", McGraw Hill Higher Education, 7th Edition, 2017.

Course Outcomes:
After the completion of the course, students will be able to
1. Explain three phase balanced and unbalanced circuits.
2. Illustrate transient response of RLC circuits for DC and AC excitations.
3. Outline the Laplace and inverse Laplace transforms.
4. Interpret various parameters for a given two-port network.
5. Illustrate the concepts of network topology
6. Translate the given electrical system into network function

COs – POs Mapping

COs	PO_1	PO_2	PSO_1
1	2	2	1
2	2	2	1
3	2	2	1
4	2	2	1
5	2	2	1
6	2	2	1

3–Strongly linked | 2–Moderately linked | 1–Weakly linked

Syllabus:

Unit I

Three Phase Circuits

Three Phase Circuits: Advantages of 3-phase system over 1-phase systems, 3-phase balanced system connections-star connection, delta connection and their comparison, Unbalanced 3-phase connections-delta connection, 3-wire star connection, 4-wire star connection, measurement of power in 3-phase circuits using 2-wattmeter method for balanced load

10 Hours

Unit II

Laplace transforms – Definition and Laplace transforms of standard functions– Shifting theorem – Transforms of derivatives and integrals, Inverse Laplace transforms and applications.

Transient Analysis- Transient response of R-L, R-C and R-L-C series circuits for DC and sinusoidal excitations – Initial conditions - Solution using differential equation and Laplace transform approach.

14 Hours

Unit III

Two-port networks

Open circuit, short circuit, transmission, Inverse transmission, hybrid and inverse hybrid parameters and their inter relations, interconnection of two port networks, T and Π representation of two-port networks.

12 Hours

Unit IV

Network Functions: Driving point impedance and admittance, Transfer impedance and admittance, current and voltage transfer ratio, Network functions of ladder and non-ladder networks, Poles and zeros of a network function

Network Topology: Terminology used in network topology, network graph and tree, basic cut-set and basic tie-set matrices for planar networks, concept of duality and dual networks.

12 Hours

Total: 48 Hours

Course Plan:

Lecture No.	Learning Objectives	Topics to be covered	Reference (s)
Unit-I: Three Phase Circuits			
1	To understand various concepts in Electrical Circuit Analysis-II	Introduction to Electrical Circuit Analysis-II	---
2	To understand the advantages of 3-phase system over 1-phase systems	Advantages of 3-phase system over 1-phase systems	T1, T2, T3
3	To understand various 3-phase balanced system connections	3-phase balanced system connections	T1, T2, T3
4	To understand star connection, delta connection and their comparison	Star connection, delta connection and their comparison	T1, T2, T3
5	To understand various Unbalanced 3-phase connections	Unbalanced 3-phase connections	T1, T2, T3
6	To understand unbalanced 3-phase delta connection	Unbalanced 3-phase delta connection	T1, T2, T3
7	To understand unbalanced 3-phase 3-wire star connection	Unbalanced 3-phase 3-wire star connection	T1, T2, T3
8	To understand unbalanced 3-phase 4-wire star connection	Unbalanced 3-phase 4-wire star connection	T1, T2, T3
9	To apply 2-wattmeter method for measurement of power in a 3-phase balanced load.	Measurement of power in 3-phase circuits using 2-wattmeter method for balanced load	T1, T2, T3
10	To apply 2-wattmeter method for measurement of power in a 3-phase balanced load.	Measurement of power in 3-phase circuits using 2-wattmeter method for balanced load	T1, T2, T3
Unit-II: Laplace transforms & Transient Analysis			

11	To understand definition and Laplace transforms of standard functions	Definition and Laplace transforms of standard functions	T1, T2, T3
12	To understand shifting theorem and transforms of derivatives and integrals	Shifting theorem – Transforms of derivatives and integrals	T1, T2, T3
13	To understand inverse Laplace transforms and applications	Inverse Laplace transforms and applications	T1, T2, T3
14	To understand the Initial conditions	Initial conditions	T1, T2, T3
15	To understand the transient response of RL series circuits for DC excitation	Transient response of RL series circuits for DC excitation	T1, T2, T3
16	To understand the transient response of RC series circuits for DC excitation	Transient response of RC series circuits for DC excitation	T1, T2, T3
17	To understand the transient response of RLC series circuits for DC excitation	Transient response of RLC series circuits for DC excitation	T1, T2, T3
18	To understand the transient response of RL series circuits for sinusoidal excitation	Transient response of RL series circuits for sinusoidal excitation	T1, T2, T3
19	To understand the transient response of RC series circuits for sinusoidal excitation	Transient response of RC series circuits for sinusoidal excitation	T1, T2, T3
20	To understand the transient response of RLC series circuits for sinusoidal excitation	Transient response of RLC series circuits for sinusoidal excitation	T1, T2, T3
21	To find the solution of a given electric circuit using differential equation approach	Solution using differential equation approach	T1, T2, T3
22	To find the solution of a given electric circuit using differential equation approach	Solution using differential equation approach	T1, T2, T3
23	To find the solution of a given electric circuit using Laplace transform approach	Solution using Laplace transform approach	T1, T2, T3
24	To find average value, R.M.S. value, form factor and peak factor for various periodic waveforms	Solution using Laplace transform approach	T1, T2, T3
Unit-III: Two-port Networks			
25	To derive impedance parameters for the given two port network	Impedance parameters	T1, T2, T3
26	To derive admittance parameters for the given two port network	Admittance parameters	T1, T2, T3
27	To derive transmission line parameters for the given two port network	Transmission line parameters	T1, T2, T3
28	To derive inverse transmission line parameters for the given two port network	Inverse transmission line parameters	T1, T2, T3
29	To derive hybrid parameters for the given two port network	Hybrid parameters	T1, T2, T3
30	To derive inverse hybrid parameters for the given two port network	Inverse hybrid parameters	T1, T2, T3
31	To outline the relation between different two-port network parameters	Inter relation between different two-port network parameters	T1, T2, T3
32	To outline the relation between different two-port network parameters	Inter relation between different two-port network parameters	T1, T2, T3
33	To outline the interconnection of two port networks	Interconnection of two port networks	T1, T2, T3
34	To outline the interconnection of two	Interconnection of two port networks	T1, T2, T3

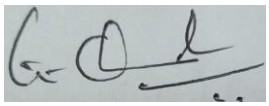
	port networks		
35	To represent the given two-port network in the form of T and Π network	T and Π representation of two-port networks	T1, T2, T3
36	To represent the given two-port network in the form of T and Π network	T and Π representation of two-port networks	T1, T2, T3
Unit-IV: Network Functions & Network Topology			
37	To determine driving point impedance and admittance of the given circuit.	Driving point impedance and admittance	T1, T2, T3
38	To determine transfer impedance and admittance of the given circuit.	Transfer impedance and admittance	T1, T2, T3
39	To determine current and voltage transfer ratios of the given circuit.	Current and voltage transfer ratio	T1, T2, T3
40	To determine network function for ladder network	Network functions of ladder and non-ladder networks	T1, T2, T3
41	To determine network function for non-ladder network	Network functions of non-ladder networks	T1, T2, T3
42	To find poles and zeros of a network function.	Poles and zeros of a network function	T1, T2, T3
43	To outline the terminology used in network topology	Terminology used in network topology	T1, T2, T3
44	To illustrate network graph and tree for the given electrical circuit	Network graph and tree	T1, T2, T3
45	To find the basic cut-set matrix for the given circuit	Basic cut-set matrix	T1, T2, T3
46	To find the basic tie-set matrix for the given circuit	Basic tie-set matrix	T1, T2, T3
47	To understand duality and dual networks	concept of duality and dual networks	T1, T2, T3
48	To understand duality and dual networks	concept of duality and dual networks	T1, T2, T3

Where T stands for Textbook

Evaluation Scheme:

Component	Duration (minutes)	Marks	% of weightage	Date & Time	Venue
Sessional Test – 1	90	40	20 (80% of marks from best one+20% of marks from other)	30-9-24 to 2-10-24 Time will be intimated later	Block-3
Sessional Test – 2	90	40		28-11-24 to 04-11-24 Time will be intimated later	Block-3
Assignment Test	120	30	10	28-11-24 to 04-11-24 Time will be intimated later	Block-3
End semester exam	180	70	70	05-12-24 to 21-12-24 Time will be intimated later	Block-3

Chamber Consultation Hour : 4:00 PM to 5:00 PM
Venue : Block 3- Second Floor EEE staff room
Notices : EEE Department -Main notice board



Dr. G.Chnadra Sekhar
Signature of the Instructor

Dr.P.Ramana
Signature of the course-coordinator